



[4658] – 603

Seat No.	
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T.E. (Information Technology) (Semester – I) Examination, 2014
COMPUTER NETWORK TECHNOLOGY
(2012 Course)

Time : 3 hours

Max. Marks : 70

Instructions : 1) Neat diagrams must be drawn **wherever** necessary.
2) Black figures to the **right** indicate **full** marks.
3) Your answers will be valued as whole.
4) Assume suitable data, if **necessary**.

1. a) Explain parameters for quality of service in network layer. 6
b) State special IP addresses and the private IP addresses. 4

OR

2. a) Calculate the following for a network address 192. 168. 1. 0127
a) No. of valid subnets
b) No. of actual hosts per subnet
c) Network and broadcast address for each subnet. 6
b) Explain TCP header format. 4
3. a) Explain how to achieve reliability at transport layer. 4
b) Explain all socket primitives used by client. 6

OR

4. a) Differentiate FTP and TFTP. 4
b) What is the purpose of SMI and MIB in relation to SNMP. 6
5. a) Explain architecture of Blue tooth. 10
b) Explain frame format of 802.16 8

OR

6. a) Describe MAC layer mechanism of IEEE 802.11 10
b) Compare Ethernet and wireless networks. 8

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7. a) List different routing protocols used by WSN with examples. **8**
b) Explain set up phase and steady state phase of LEACH protocol. **8**
OR
8. a) List any six applications of sensor network. **8**
b) Explain following terms. **8**
i) Data aggregation
ii) Data diffusion
iii) Data dissemination
9. a) Discuss different addresses and identifiers in WSN. **8**
b) Explain Geographic routing in WSN. **8**
OR
10. a) Write short notes on (**any 2**). **16**
i) DSR
ii) BYOD
iii) Content Based routing.
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